ON SYMMETRIC FUNCTIONS AND SEMINVARIANTS (continued)

[932, pp. 304–328]

55. We can, by means of the umbral notation, write down for the blunt seminvariants of a given weight (indefinite forms, not the above-mentioned specific forms) expressions far more simple than those which are given by the foregoing theories: we can, in fact, find without difficulty monomial umbral expressions; and in many cases obtain also the sharp forms. To illustrate this, I consider the weight 10: I write down for convenience the symbols of the sharp forms (though the knowledge of these is in nowise required) and I form a table as follows:

Sharp forms, finals in AO.		
$k\infty f^2$	1	$(\alpha-\beta)^{10}$
$ci\infty ce^2$	2	$(\alpha-\beta)^8(\alpha-\gamma)^2$
$dh\infty b^2e^2$	3	$(\alpha-\beta)^8(\alpha-\gamma)(\alpha-\delta)$
$eg\infty bd^3$	4	$(\alpha-\beta)^6(\alpha-\gamma)^3(\alpha-\delta)$
$f^2\infty c^3d^2$	5	$(\alpha-\beta)^6(\alpha-\gamma)^2(\alpha-\delta)^2$
$c^2g\infty b^2cd$	6	$(\alpha-\beta)^6(\alpha-\gamma)^2(\alpha-\delta)(\alpha-\epsilon)$
$cdf\infty b^4d^2$	7	$(\alpha-\beta)^4(\alpha-\gamma)^2(\alpha-\delta)^2(\alpha-\epsilon)^2$
$cdf\infty b^3cf$	8	$(\alpha-\beta)^4(\alpha-\gamma)(\alpha-\delta)(\alpha-\epsilon)(\alpha-\zeta)$
$ce^2\infty c^5$	9	$(\alpha-\beta)^4(\alpha-\gamma)^2(\alpha-\delta)(\alpha-\epsilon)(\alpha-\zeta)$
$d^2e\infty b^5d^2$	10	$(\alpha-\beta)^4(\alpha-\gamma)^2(\alpha-\delta)(\alpha-\epsilon)(\alpha-\zeta)(\alpha-\eta)$
	11	$(\alpha-\beta)^4(\alpha-\gamma)(\alpha-\delta)(\alpha-\epsilon)(\alpha-\zeta)(\alpha-\eta)(\alpha-\theta)$
	12	$(\alpha-\beta)^2(\alpha-\gamma)(\alpha-\delta)(\alpha-\epsilon)(\alpha-\zeta)(\alpha-\eta)(\alpha-\theta)(\alpha-\iota)(\alpha-\kappa)$

It will be observed that all the differences used are $\alpha-\beta$, $\alpha-\gamma$, ..., containing each of them an α; hence in all the forms we have $\alpha^{10}, = k$; in $(\alpha-\beta)^{10}$, the lowest term (in AO) is $\alpha^5\beta^5, = f^2$; in $(\alpha-\beta)^8(\alpha-\gamma)^2$, the lowest term is $\alpha^4\beta^4\gamma^2, = ce^2$; and so on, viz. in each case the lowest term is the final term of the sharp form set down in the same line.

56. The form $(\alpha-\beta)^{10}$ gives at once the sharp form $k\infty f^2, = 252\alpha^5\beta^5$; we thus develop it:

$$\alpha^{10} \quad -10\alpha^9\beta \quad +45\alpha^8\beta^2 \quad -120\alpha^7\beta^3 \quad +210\alpha^6\beta^4 \quad -252\alpha^5\beta^5 \quad +210\alpha^4\beta^6 \quad -120\alpha^3\beta^7 \quad +45\alpha^2\beta^8 \quad -10\alpha\beta^9 \quad +\beta^{10}.$$

$(\alpha-\beta)^8(\alpha-\gamma)^2$ contains a term $\alpha^{10}, = k$ and thus gives a blunt form $k\infty ce^2$; if instead of it we employ the form $(\alpha-\beta)^8(\alpha-\gamma)(\beta-\gamma)$, then here as before the lowest term is $\alpha^4\beta^4\cdot\gamma^2, = ce^2$, but there is no term α^{10}: there is a term $\alpha^9\beta, = bj$, but as this

cannot appear, we must have terms of this form destroying each other. The simplest mode of effecting the development is to write

$$(\alpha-\beta)^8(\alpha-\gamma)(\beta-\gamma) = (\alpha-\beta)^8\{\alpha\beta-\gamma(\alpha+\beta)+\gamma^2\};$$

we may herein put at once $\gamma = b$, $\gamma^2 = c$, and thus the form is

$$(\alpha-\beta)^8\{\alpha\beta-b(\alpha+\beta)+c\};$$

I develop thus:

[Table: development of $(\alpha-\beta)^8\{\alpha\beta-b(\alpha+\beta)+c\}$. The four rows of binomial expansions corresponding to the factors $(\alpha-\beta)^8$, $\alpha\beta$, $-b(\alpha+\beta)$ and $+c$ are written out across ten columns of $\alpha^i\beta^{10-i}$ powers, with the row of coefficients $1, -8, +28, -56, +70, -56, +28, -8, +1$ for $(\alpha-\beta)^8$ and the corresponding shifted/scaled rows for the other three factors. Below the rule, the per-monomial sums are tabulated in twelve rows labelled bj, ci, dh, eg, f^2, b^4h, bcg, bdf, b^2cf, c^2g, cdf, ce^2 with column-wise contributions and the final divisor-column $\div -14$ giving the final integer coefficients of the sharp seminvariant $ci\infty ce^2$. A trailing tally ± 23 summarises the result.]

which, in fact, exhibits the calculation of the sharp form $ci\infty ce^2$. The disappearance of the term in bj will be noticed.

Instead of $(\alpha-\beta)^8(\alpha-\gamma)(\alpha-\delta)$ which contains α^{10}, that is, k, we may take

$$(\alpha-\beta)^8(\gamma-\delta)^2,$$

that is,

$$(i-8bh+28cg-56df+35e^2)(c-b^2):$$

this is $ci\infty b^2e^2$, a blunt form; by subtracting from it $ci\infty ce^2$, we could obtain the next sharp form $dh\infty b^2e^2$; but this in passing; it does not appear that there is any monomial umbral expression for the last-mentioned form.

I do not stop to examine the next following forms, but pass on at once to the last of them; instead of the expression given, we may take the expression

$$(\alpha-\beta)^2(\gamma-\delta)^2(\epsilon-\zeta)^2(\eta-\theta)^2(\iota-\kappa)^2,$$

that is, $(c-b^2)^5$, which is in fact the sharp form $c^5\infty b^{10}$.

Seminvariants of a given Degree: Generating Functions. Art. Nos. 57 to 59.

57. We may consider the seminvariants of a given degree, and arrange them according to their weights: thus in each case writing down the series of finals, and for a reason that will appear also the conjugates of these finals (see Table of Conjugates, ante No. 54).

For degree 2, or quadric seminvariants, we have

2	3	4	5	6 ...
C, b^2	–	C^2, c^2	–	C^3, d^2

there is here for every even weight (beginning with 2) a single form, and for every odd weight no form: the number of forms of the weight w is thus = coeff. of x^w in $x^2\div(1-x^2)$, or writing for shortness 2 to denote $1-x^2$ (and similarly $3,4,\ldots$ to denote $1-x^3$, $1-x^4$, $\ldots$), say that for degree 2, Generating Function, G.F., is $=x^2\div 2$.

For degree 3, or cubic seminvariants, we have

3	4	5	6	7 ...
D, b^3	–	CD, bc^2	D^2, b^2d	C^2D, bc^3

the counting is most easily effected by means of the conjugate forms; these contain all of them the factor D, and omitting this factor we have all the combinations of C, D which make up the weight $w-3$, viz. for weight w, we have number of ways in which $w-3$ can be made up with the parts 2, 3: that is,

$$\text{for degree 3, G.F. is } = x^3\div 2\cdot 3.$$

Similarly for degree 4 or quartic seminvariants, we have terms each containing E, and removing this factor, we have all the combinations of C, D, E which make up the weight $w-4$, viz.

$$\text{for degree 4, G.F. is } = x^4\div 2\cdot 3\cdot 4.$$

Thus for degrees

$$2,\ 3,\ 4,\ 5,\ 6,\ \ldots$$

the G.F.'s are

$$= x^2 \div 2,\ x^3 \div 2\cdot 3,\ x^4 \div 2\cdot 3\cdot 4,\ x^5 \div 2\cdot 3\cdot 4\cdot 5,\ x^6 \div 2\cdot 3\cdot 4\cdot 5\cdot 6,\ \ldots$$

58. We may analyse these results by separating the finals into classes. I use the expression $b, c, d, \ldots$ are discrete letters, meaning thereby that they are distinct letters, not of necessity consecutive but with any intervals between them. Thus deg. 3, if (b, c) are discrete letters, then the finals are b^3, and bc^2; deg. 4, if b, c, d are discrete letters, then the finals are b^4, bc^3, b^2c^2, and bcd^2; and so on, the number of classes being doubled at each step, as will presently appear for the weights 5 and 6 respectively.

I notice also a property of the conjugates of these classes; for b^3 and bc^2 themselves the conjugates are D and CD, and these occur as factors, D in the conjugate of every form of the class b^3 (for instance conjugates of c^3, d^3 are D^2, D^3), and CD in the conjugate of every form of the class bc^2 (for instance, the conjugates of bd^2, ce^2 are C^2D, C^2D^2); and the like in other cases, viz. for any class whatever the conjugate of the first or representative form occurs as a factor in the conjugates of the several other forms belonging to the same class.

59. With these explanations, the expressions for the several G.F.'s are obtained without difficulty, and we have

$$\begin{array}{lll} \text{deg. } 2, & \text{class } C, b^2 & \text{G.F.} = x^2 \div 2, \\ \text{deg. } 3, & \quad ,,\ D, b^3 & \quad ,,\ = x^3 \div 3, \\ & \quad ,,\ CD, bc^2 & \quad ,,\ = x^5 \div 2\cdot 3; \end{array}$$

we ought here to have

$$x^3 \div 2\cdot 3 = x^3 \div 3 + x^5 \div 2\cdot 3, \text{ viz. in verification}$$

$$x^3 = x^3\cdot 2 = x^3 - x^5.$$

$$\begin{array}{lll} \text{deg. } 4, & \text{class } E, b^4 & \text{G.F.} = x^4 \div 4, \\ & \quad ,,\ DE, bc^3 & \quad ,,\ = x^7 \div 3\cdot 4, \\ & \quad ,,\ CE, b^2c^2 & \quad ,,\ = x^6 \div 2\cdot 4, \\ & \quad ,,\ CDE, bcd^2 & \quad ,,\ = x^9 \div 2\cdot 3\cdot 4; \end{array}$$

we ought here to have $x^4 \div 2\cdot 3\cdot 4 = x^4 \div 4 + x^7 \div 3\cdot 4 + x^6 \div 2\cdot 4 + x^9 \div 2\cdot 3\cdot 4$, viz. in verification

$$x^4 = x^4\cdot 2\cdot 3 = x^4 - x^6 - x^7 + x^9.$$

deg. 5,	class F, b^5	G.F. $= x^5 \div 5,$
	„ EF, bc^4	„ $= x^9 \div 4 \cdot 5,$
	„ DF, b^2c^3	„ $= x^8 \div 3 \cdot 5,$
	„ CF, b^3c^2	„ $= x^7 \div 2 \cdot 5,$
	„ DEF, bc^2d^2	„ $= x^{12} \div 3 \cdot 4 \cdot 5,$
	„ CEF, b^2cd^2	„ $= x^{11} \div 2 \cdot 4 \cdot 5,$
	„ CDF, b^2c^3	„ $= x^{10} \div 2 \cdot 3 \cdot 5,$
	„ $CDEF, bcde^2$	„ $= x^{14} \div 2 \cdot 3 \cdot 4 \cdot 5;$

and for the sum of the eight terms

$$\text{G.F.} = x^5 \div 2 \cdot 3 \cdot 4 \cdot 5,$$

which may be verified as before.

deg. 6,	class G, b^6	G.F. $= x^6 \div 6,$
	„ FG, bc^5	„ $= x^{11} \div 5 \cdot 6,$
	„ EG, b^2c^4	„ $= x^{10} \div 4 \cdot 6,$
	„ DG, b^3c^3	„ $= x^9 \div 3 \cdot 6,$
	„ CG, b^4c^2	„ $= x^8 \div 2 \cdot 6,$
	„ EFG, bc^3d^2	„ $= x^{15} \div 4 \cdot 5 \cdot 6,$
	„ $DFG, b^2c^2d^2$	„ $= x^{14} \div 3 \cdot 5 \cdot 6,$
	„ CFG, b^3cd^2	„ $= x^{13} \div 2 \cdot 5 \cdot 6,$
	„ DEG, b^2cd^3	„ $= x^{13} \div 3 \cdot 4 \cdot 6,$
	„ $CEG, b^2c^2d^2$	„ $= x^{12} \div 2 \cdot 4 \cdot 6,$
	„ CDG, b^3cd^2	„ $= x^{11} \div 2 \cdot 3 \cdot 6,$
	„ $DEFG, bcde^3$	„ $= x^{18} \div 3 \cdot 4 \cdot 5 \cdot 6,$
	„ $CEFG, b^2cde^2$	„ $= x^{16} \div 2 \cdot 4 \cdot 5 \cdot 6,$
	„ $CDFG, b^2c^2de^2$	„ $= x^{15} \div 2 \cdot 3 \cdot 5 \cdot 6,$
	„ $CDEG, bcde^3$	„ $= x^{15} \div 2 \cdot 3 \cdot 4 \cdot 6,$
	„ $CDEFG, bcdef^2$	„ $= x^{20} \div 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6;$

and for the sum of the sixteen terms

$$\text{G.F.} = x^6 \div 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6,$$

which may be verified as before.

Reducible Seminvariants—Perpetuants. Art. Nos. 60 to 64.

60. Seminvariants of the degrees 2 and 3 are irreducible, or say they are perpetuants. Hence by what precedes, as regards perpetuants

$$\text{for degree 2, G.F.} = x^2 \div 2;$$

$$\text{for degree 3, G.F.} = x^3 \div 2 \cdot 3.$$

For the degree 4 (if as before b, c, d denote discrete letters), then the finals are b^4, bc^3, b^2c^2 and bcd^2. For a final $= b^2 \cdot b^2$ or $b^2c^2 = b^2 \cdot c^2$, we have evidently a product of two quadric seminvariants ending in b^2 and b^2, or in b^2 and c^2, with the same final term as the quartic seminvariant; so that, considering the quartic seminvariants arranged with their finals in AO, and adding to such quartic seminvariant a proper numerical multiple of the product in question, we obtain a quartic seminvariant the final term whereof is in AO higher than the original final term b^4 or b^2c^2, and such quartic seminvariant is thus said to be reducible; a quartic seminvariant not thus reducible is a perpetuant. The quartic perpetuants are consequently those which end in bc^3 or bcd^2. The lowest form is that ending in bc^3, of the weight 7. Taking the sum of the G.F.'s for the forms bc^3 and bcd^2 respectively, the G.F. for quartic perpetuants is

$$x^7 \div 3 \cdot 4 + x^9 \div 2 \cdot 3 \cdot 4,$$

viz. this is

$$x^7(1 - x^2) + x^9 \div 2 \cdot 3 \cdot 4,$$

or finally

$$\text{G.F.} = x^7 \div 2 \cdot 3 \cdot 4.$$

As an instance of a reduction, we have

$$(d^2 \infty b^2c^2) - (c \infty b^2)(e \infty c^2) = (ce \infty c^3),$$

viz. this is

$$(d \infty b^2c^2) = (c - b^2)(e - 4bd + 3c^2) - (ce - d^2 - b^2e + 3bcd - c^3).$$

We have also

$$(d^2 \infty b^2c^2) = (d \infty b^3)^2 + 4(c \infty b^2)^3,$$

viz.

$$(d \infty b^2c^2) = (d - 3bc + 2b^3)^2 + 4(c - b^2)^3,$$

but this is not a reduction, there are on the right-hand side terms of the degree 6, which is higher than the degree of the seminvariant $d^2 \infty b^2c^2$. In general, we say that a seminvariant of any given degree is reducible when we can, by adding to it products of its own degree of seminvariants of inferior degrees, reduce it to a seminvariant the final of which is in AO higher than the original final.

61. For the degree 5 (taking b, c, d, e to denote discrete letters), if the final be b^5, bc^4, b^2c^3, b^3c^2, bc^2d^2 or b^2cd^2, then the seminvariant will be reducible; a perpetuant must have therefore a final bcd^3 or $bcde^2$. But it is not true that every quintic seminvariant with either of these finals is a perpetuant. To explain this, observe that the first mentioned six finals are some of them in one way only, some of them in two ways, expressible as a product of power-enders, or say they are singly, or else doubly, composite: viz. we have

$$b^5 = b^2 \cdot b^3;\ bc^4 = c^2 \cdot bc^2;\ b^2c^3 = b^2 \cdot c^3;\ b^3c^2 = c^2 \cdot b^3 = b^2 \cdot bc^2;$$

$$bc^2d^2 = c^2 \cdot bd^2 = d^2 \cdot bc^2;\ b^2cd^2 = b^2 \cdot cd^2.$$

For a doubly composite form, for instance b^3c^2, forming first the product of the quadric and cubic seminvariants ending in c^2, b^3 respectively, and secondly the product of the quadric and cubic seminvariants ending in b^2 and bc^2 respectively, we have two products each with the final b^3c^2, and forming a linear combination so as to eliminate this term b^3c^2, we have thus it may be a quintic seminvariant with a final such as bcd^3 or $bcde^2$, and the process then furnishes a reduction of such a quintic seminvariant. Or on the other hand, it may be that the finals of the degree 5 all of them disappear, and we have a relation between products of the form in question (i.e. of a quadric and a cubic seminvariant) and seminvariants of a degree inferior to 5, say this is a quintic syzygy.

In particular, a non-composite final first presents itself for the weight 12, viz. here the finals are bce^2, bcd^3, c^3d^2, the last of these is doubly composite, and it furnishes a reduction of bcd^3. For the weight 13, the finals are b^3f^2, b^2de^2, bc^2e^2, bd^4, c^4d^3 which are each of them singly or doubly composite; for the weight 14, they are b^2cf^2, b^4c^3, $bcde^2$, c^3e^2 and cd^4, and here the doubly composite form furnishes a reduction of $bcde^2$. For the weight 15, we have a final bce^3 which gives a quintic perpetuant. I have, in fact, in my paper "A Memoir on Seminvariants," *American Journal of Mathematics*, vol. VII. (1885), pp. 1–25, [828], worked out the theory of quintic syzygies and perpetuants, and subsequently connecting this with the present theory of finals, I succeeded in showing that, when the doubly composite final contains a b, then there is not a reduction but a syzygy; we thus have

$$\text{G.F. for finals } b^3c^2, b^3d^2, \ldots = x^7 \div 2,$$

whence for the two forms

$$\text{,, } bc^2d^2, \ldots = x^{11} \div 2 \cdot 4,$$

$$\text{G.F. is } x^7 \div 2 + x^{11} \div 2 \cdot 4 = \{x^7(1 - x^4) + x^{11}\} \div 2 \cdot 4,$$

or say for S_5, the number of quintic syzygies G.F. is $= x^7 \div 2 \cdot 4$.

I further satisfied myself that the finals for the quintic perpetuants are $bc0e^3$, $bc0ef^2$, viz. the b, c, e, f being discrete letters, the interposed 0 denotes that c and e are not consecutive letters. The conjugates of these forms contain the factors D^2EF and CD^2EF respectively, and it hence appears that the G.F.'s are $= x^{15} \div 3 \cdot 4 \cdot 5$ and $x^{17} \div 2 \cdot 3 \cdot 4 \cdot 5$; adding these, we find

$$\text{for quintic perpetuants G.F. is } = x^{15} \div 2 \cdot 3 \cdot 4 \cdot 5,$$

which expression was given in the memoir just referred to: the result was obtained by investigating in the first instance an expression for S_5, the number of quintic syzygies of a given weight. The course of Stroh's investigation to be presently given is different; he determines directly the number of perpetuants, and we may if we please use conversely this result to obtain the number of syzygies.

62. The foregoing theory of reduction is independent of the form of the seminvariants, which may be blunt or sharp at pleasure: the actual formulas will of course be different, and they are very much more simple for the sharp seminvariants, viz. here in many cases a seminvariant is found to be equal to a product of seminvariants of inferior degrees. I subjoin the following table of the reduction of the several sharp seminvariants up to the weight 12; the forms referred to are the tabulated forms, and to mark that this is so I write down in each case the numerical coefficients of the initial and final terms, viz. instead of $c\infty b^2$, $d\infty b^3$, &c., I write $c\infty - b^2$, $d\infty 2b^3$, &c. As appears by the table, these are for shortness denoted by C, D respectively, and so for weight 4, the forms are called E, E_2, for weight 5, F, F_2, for weight 6, G, G_2, G_3, G_4, and so on, the unsuffixed letters having thus an implied suffix, not 0 but 1. The table is

Table of Reductions.

[Table: large four-column reduction table giving for each weight $w = 2, 3, \ldots, 12$ the sharp seminvariants with their initial and final terms and the corresponding reduction expressions in capital-letter seminvariants $C, D, E, E_2, F, F_2, G, G_2, G_3, G_4, H, H_2, \ldots$; the precise numerical coefficients and product expressions are exhibited line-by-line in the printed scan and are too dense to faithfully transcribe in linear form here. The columns are headed by the weight w, and within each column the entries are arranged as: seminvariant in the form "(final-term) ∞ (initial-term)", followed by its reduction (where present) as a polynomial in $C, D, E, \ldots$. Entries without a reduction are perpetuants. Headings for the four blocks are $w = 2, 3, 4, 5, 6, 7, 8, 9, 10$ in the left two columns and $w = 11, 12$ in the right two columns. Representative entries: $w = 2$: $c\infty - b^2$, C; $w = 3$: $d\infty 2b^3$, D; $w = 4$: $e\infty 3c^2$, E, with $E_2 = C^2$; $c^2\infty b^4$, F; $w = 5$: $f\infty - bc^2$, F, $F_2 = CD$; $w = 6$: $g\infty - 10d^2$, G; $G_2 = CE - G_2$; G_3, $G_4 = C^3$; etc. Where no reduction is given, the form is irreducible, i.e. it is a perpetuant.]

Where no reduction is given, the form is irreducible, i.e. it is a perpetuant.

63. As to these reductions, it may be observed that in very many cases we have the sharp seminvariant given as an actual product $E_2 = C^2$, $F_2 = CD$, $G_4 = C^3$, &c. We have next other reductions such as $G_3 = CE - G_2$, where on the right-hand side there is a single product; this has a final the same as that of the seminvariant which is to be reduced, so that, eliminating this term from the seminvariant and product in question, we have an expression which must be a linear combination (with numerical coefficients) of the preceding seminvariants of the same weight. To take a less simple example, $L_5 = -DI + L_3 + 2L_4$; here $L_5 = -fg + 16c^2h \ldots - 70b^3e^2$, and $DI = (d - 3bc + 2b^3)(i \ldots + 35e^2)$ has the final $+70b^3e^2$. The verification is

$$\begin{array}{rllll} -DI = & -di & & \ldots & -70b^3e^2 \\ +L_3 = & di & -2eh + fg & & \\ -2L_4 = & & 2eh - 2fg & & \\ \hline L_5 = & & & -fg & -70b^3e^2. \end{array}$$

The only case in which we have on the right-hand side two products is $(d^2g \infty bcd^3)$, $M_9 = -3GG_2 + 5EI_2 - 2M_7$; viz. here the final of M_9 is bcd^3 which is incomposite (viz. it is not the product of two power-enders), this is in fact the first instance of a quintic seminvariant with an incomposite final and which is nevertheless reducible. For observe, the next seminvariant M_{10} has the final c^3d^2, which is a product in the two ways $c^2 \cdot cd^2$ and $c^3 \cdot d^2$; we have thus the two products $(e \infty c^2)(eg \infty cd^2)$ and $(ce \infty c^3)(g \infty d^2)$, that is, EI_2 and GG_2 with the same final c^3d^2, and combining them so as to eliminate this term we have an expression having the final bcd^3, and which is thus expressible in terms of M_9 and preceding seminvariants: the verification is

$$\begin{array}{rllll} -3GG_2 = & -3ceg + 3d^2g & & \ldots & +60bcd^3 - 30c^3d^2 \\ +5EI_2 = & +5ceg & -40d^2g + \ldots & & \\ -2M_7 = & -2ceg + 2cf^2 & & & \\ \hline & & 2cf^2 + 5d^2g \ldots & & + \ldots . \end{array}$$

64. I annex to this a table (taken from the square diagrams) for the initials and finals of the sharp seminvariants for the weights 13, 14, 15, and 16.

[Table: large four-column listing of sharp seminvariants for the weights $w = 13, 14, 15, 16$, organized in four parallel columns each headed by the weight and by the capital-letter symbol assigned to that weight (N, O, P, Q respectively). For each weight the entries are pairs (final-term, initial-term) printed line-by-line, with running numerical labels $1, 2, 3, \ldots$ down the right-hand side of each column. The entries cover all sharp seminvariants of the given weight. For $w = 13$: $n \infty bdf^2$, N, with subsequent entries including $dk \infty bcf^2$, $fi \infty bce^3$, $ej \infty bd^2e^2$, $\ldots$, terminating with $c^5d \infty b^{15}$. For $w = 14$: $o \infty h^2$, O, with subsequent entries including $cm \infty cg^2$, $ek \infty ef^2$, $dl \infty b^2g^2$, $fj \infty bdf^2$, $\ldots$, terminating with $c^7 \infty b^{14}$. For $w = 15$: $p \infty bg^2$, P, with subsequent entries including $2cm \infty bg^2$, $bo \infty bfg^2$, $edl \infty bcg^2$, $fk \infty bef^2$, $cn \infty bce^3$, $ej \infty bd^2g$, $gk \infty c^2f^2$, $f^2i \infty c^2d^3$, $\ldots$, terminating with $c^7d \infty b^{15}$. For $w = 16$: $q \infty c^4$, Q, with subsequent entries including $eo \infty cf^2$, $cm \infty bdg^2$, $fl \infty bcg^2$, $cek \infty bcfg^2$, $dn \infty b^2h^2$, $i^2 \infty d^2f^2$, $ch^2 \infty bc^2g^2$, $cg^2 \infty bd^5$, $efh \infty b^2ce^3$, $\ldots$, terminating with $c^8 \infty b^{16}$. Within each column the running tally on the right-hand side increases as the list descends. The precise pair-by-pair transcription is rendered ambiguous by the density and partial-illegibility of the scan.]

It would be interesting to complete this into a table of reductions as given for the weights 2 to 12.

The Strohian Theory Resumed: Application to Perpetuants. Art. Nos. 65 to 71.

65. We can by means hereof establish, in regard to the specific blunt seminvariants, a general theory of reduction, or say a theory of the relations which exist between the seminvariants of a given degree and the powers and products of seminvariants of inferior degrees. To exhibit the form of these, it will be sufficient to take Ω a sum of two parts, $= \Omega' + \Omega''$, but the more general assumption is Ω a sum of any number of parts, $= \Omega' + \Omega'' + \Omega''' + \ldots$. Taking then $\Omega = \Omega' + \Omega''$, where for the Ω' and Ω'' separately the sum of the $(x, y, z, \ldots)$ is $= 0$, suppose that to the $(0, C, D, E, \ldots)$ of Ω there correspond $(0, C', D', E', \ldots)$ for Ω' and $(0, C'', D'', E'', \ldots)$ for Ω''. We have

$$C = C' + C'',$$

$$D = D' + D'',$$

$$E = E' + E'' + C'C'',$$

$$F = F' + F'' + C'D'' + C''D',$$

$$G = G' + G'' + C'E'' + C''E' + D'D'',$$

the law of which is obvious.

66. We have, for instance,

$$\Omega^4 = (\Omega' + \Omega'')^4 = \Omega'^4 + 6\Omega'^2\Omega''^2 + \Omega''^4, \text{ (since } \Omega' = 0,\ \Omega'' = 0),$$

that is,

$$\begin{gathered}(C' + C'')^2 \cdot c^2 = C'^2 \cdot c^2 + 6C' \cdot b^2 \cdot C'' \cdot b^2 + C''^2 \cdot c^2 \\ +(E' + E'' + C'C'') \cdot b^4 = E' \cdot b^4 + E'' \cdot b^4\end{gathered}$$

where, and in what follows, c^2, b^4, b^2 are for shortness written instead of $[c^2], [b^4], [b^2]$ to denote the specific blunt seminvariants ending in c^2, b^4, b^2 respectively.

The terms in C'^2, C''^2, E', E'' are identical on each side of the equation and destroy each other; omitting these, we have only the terms in $C'C''$ which must be equivalent on the two sides of the equation, and comparing coefficients we find the relation

$$2c^2 + b^4 = 6 \cdot b^2 \cdot b^2,$$

which of course means $2[c^2] + [b^4] = 6[b^2][b^2]$, viz. this is

$$2(2c - 8bd + 6c^2) + (-4e + 16bd + 12c^2 - 48b^2c + 24b^4) = 6(-2c + 2b^2)^2.$$

In like manner, for $\Omega^6 = (\Omega' + \Omega'')^6$, we have

$$\begin{gathered}(C' + C'')^3 \cdot d^2 \\ +(D' + D'')^2 \cdot c^3 \\ +(C' + C'')(E' + E'' + C'C'') \cdot b^2c^2 \\ +(G' + G'' + C'E'' + C''E' + D'D'') \cdot b^6\end{gathered}$$

equal to

$$\{C'^3 \cdot d^2\} + 15\{C'^2 \cdot c^2\}\{C'' \cdot b^2\} + 20D' \cdot b^3 \cdot D'' \cdot b^3 + 15\{C'^2 \cdot b^2\}\{C''^2 \cdot c^2\} + \{C''^3 \cdot d^2\} + \{D''^2 \cdot c^3\} + (D'^2 \cdot c^3) + \{G'' \cdot b^6\} + \{C'E'' \cdot b^2c^2\}.$$

Here omitting the terms which destroy each other and comparing the coefficients of the remaining terms, viz. $C'^2C'' + C''^2C'$, $D'D''$ and $C'E'' + C''E'$, we find the relations

$$3d^2 + b^2c^2 = 15 \cdot c^2 \cdot b^2,$$

$$2c^3 + b^6 = 20 \cdot b^3 \cdot b^3,$$

$$b^2c^2 + b^6 = 15 \cdot b^2 \cdot b^4,$$

which may be easily verified. There are on the right-hand side only products of two parts, but this is on account of the special assumption $\Omega = \Omega' + \Omega''$, a sum of two parts.

67. I write now

$$\Omega_2 = \alpha x + \beta y, \quad S_2x = 0,$$

$$\Omega_3 = \alpha x + \beta y + \gamma z, \quad S_3x = 0,$$

$$\Omega_4 = \alpha x + \beta y + \gamma z + \delta w, \quad S_4x = 0,$$

$$\Omega_5 = \alpha x + \beta y + \gamma z + \delta w + \epsilon t, \quad S_5x = 0,$$

$$\Omega_6 = \alpha x + \beta y + \gamma z + \delta w + \epsilon t + \zeta u, \quad S_6x = 0,$$

and I say that Ω_2 and Ω_3 cannot break up: but that Ω_4 breaks up if it becomes a sum of $2 + 2$ terms (i.e. a sum of two parts Ω_2 for each of which $S_2x = 0$, and so in other cases): that Ω_5 breaks up if it becomes a sum of $2 + 3$ terms, Ω_6 breaks up if it becomes a sum of $2 + 4$ or $2 + 2 + 2$ terms, or if it becomes a sum of $3 + 3$ terms: and similarly for any higher suffix.

The condition that Ω_4 may break up is $x + y = 0$, $x + z = 0$, or $y + z = 0$, or what is the same thing it is $\Pi_3(x + y) = 0$, where $\Pi_3(x + y)$ is the product of the three sums each containing x; this is a symmetric function, we in fact have

$$\Pi_3(x + y) = x^3 + x^2(y + z + w) + x(yz + yw + zw) + yzw, \ = xyz + xyw + xzw + yzw, \ = -D.$$

The condition in order that Ω_5 may break up is $x + y = 0, \ldots,$ or $w + t = 0$, say this is $\Pi_{10}(x + y) = 0$, where $\Pi_{10}(x + y)$ denotes the product of the ten sums $x + y, \ldots, w + t$. It will be shown that we have $\Pi_{10}(x + y) = -D^2E + CDF - F^2$.

The condition in order that Ω_6 may break up is, $x + y = 0, \ldots,$ or $t + u = 0$, or again if $x + y + z = 0$, $\ldots,$ or $x + t + u = 0$, viz. it is $\Pi_{15}(x + y)\Pi_{10}(x + y + z) = 0$, where $\Pi_{15}(x + y)$ is the product of the fifteen sums $x + y, \ldots, t + u$, and $\Pi_{10}(x + y + z)$ is the product of the ten sums $x + y + z, \ldots, x + t + u$, each containing x: $\Pi_{15}(x + y)$ and

$\Pi_{10}(x+y+z)$ are symmetric functions, the expressions for which will be given further on: the weights in the capital letters are 15 and 10 respectively. And similarly for Ω with any higher suffix, we have the condition that this may break up.

I introduce the factors $\Pi_4 x = E$, $\Pi_5 x = -F$, $\Pi_6 x = G$, ... respectively and write for

$$\Omega_4 \;\; M_7 = \Pi_4 x \Pi_3(x+y) = -DE \text{ as above,}$$

$$\Omega_5 \;\; M_{15} = \Pi_5 x \Pi_{10}(x+y) = -F(-D^2E + CDF - F^2) \text{ as above,}$$

$$\Omega_6 \;\; M_{31} = \Pi_6 x \Pi_{15}(x+y)\Pi_{10}(x+y+z),$$

where observe that, for the even suffixes of Ω, the last factors $\Pi_3(x+y), \Pi_{10}(x+y+z), \ldots$ denote the products of the sums $x+y, x+y+z, \ldots$ which contain x, that is in each case the products of only half the whole number of such linear factors. The suffixes of M show the weights in the capital letters $C, D, E, F, G, \ldots$ viz. these are $4+3, =7$, $5+10, =15$, $6+15+10, =31$, and so on; the law is obvious, and for Ω_n the weight is $= 2^{n-1}-1$.

68. To explain the Strohian theory of perpetuants, I assume explicitly as presently appears. For perpetuants of any given degree δ, we consider in $\Omega^{\delta w}$ ($w = \delta$ at least) the terms containing seminvariants of the given degree: for instance when $\delta = 4$, $w = 12$, these are

$$\begin{gathered} C^4E \cdot b^2f^2 \\ +C^2DE \cdot bde^2 \\ +C^2E^2 \cdot c^2e^2 \\ +E^3 \cdot d^4, \end{gathered}$$

where the capital expressions all contain as factor the letter E of the weight 4. By making Ω to break up, it is assumed that we obtain all the reductions of the seminvariants of the degree and weight in question; and every such seminvariant, if it be reducible, will be reduced by means of the resulting formulæ. Now there are seminvariants which are not reducible by these formulæ: in the example just considered, the seminvariant bde^2 has the coefficient CD^2E containing the factor DE, $= xyzw(x+y)(x+z)(x+w)$ which vanishes when Ω_4 breaks up; so that, supposing Ω_4 to break up, the seminvariant bde^2 disappears from the formulæ, and we have no reduction of this seminvariant. And again it is assumed that every seminvariant which does not in this way disappear from the equation is reducible. The irreducible seminvariants are thus the seminvariants which, when Ω breaks up into a sum of two or more parts, disappear from the formulæ; viz. the seminvariants which thus disappear are the perpetuants.

69. In the case considered of quartic seminvariants, it has just been seen that, for the weight 12, bde^2 is a perpetuant; and so in general for the weight w, every quartic seminvariant, multiplied into a product of capitals which contains the factor DE, is a perpetuant: for the weight 7 the only term is $DE \cdot bc^3$, viz. the product

of capitals is here $= DE$; and for any higher weight w we have products which are equal to DE multiplied into products of the weight $w-7$ in C, D, E: and we thus see that the G.F. for quartic perpetuants is $= x^7 \div 2 \cdot 3 \cdot 4$.

70. For quintic perpetuants, we consider in Ω^{5w} ($w = 5$ at least) the terms which contain quintic perpetuants; for instance, when $w = 15$, the terms are

$$\begin{gathered} C^5F \cdot b^3g^2 \\ +C^2D^2F \cdot b^2df^2 \\ +C^3EF \cdot bc^2f^2 \\ +D^2EF \cdot bce^3 \\ +CE^2F \cdot bd^2e^2 \\ +CDF^2 \cdot c^2de^2 \\ +F^3 \cdot d^5, \end{gathered}$$

where the functions of the capitals all contain the factor F; the finals b^3g^2, b^2df^2, ... are arranged in AO. Supposing Ω_5 to break up, we have an expression $M, = -D^2EF + CDF^2 - F^3$, which is $= 0$, and using this value of M to eliminate the term D^2EF which belongs to the seminvariant bce^3, the final whereof is highest in AO, viz. writing $D^2EF = -M + CDF^2 - F^3$, the expression is

$$\begin{array}{lcl} C^5F \cdot b^3g^2 & \text{that is} & C^5F \cdot b^3g^2 \\ +C^2D^2F \cdot b^2df^2 & & +C^2D^2F \cdot b^2df^2 \\ +C^3EF \cdot bc^2f^2 & & +C^3EF \cdot bc^2f^2 \\ +(-M + CDF^2 - F^3) \cdot bce^3 & & -M \cdot bce^3 \\ +CE^2F \cdot bd^2e^2 & & +CE^2F \cdot bd^2e^2 \\ +CDF^2 \cdot c^2de^2 & & +CDF^2 \cdot (c^2de^2 + bce^3) \\ +F^3 \cdot d^5 & & +F^3 \cdot (d^5 - bce^3); \end{array}$$

and here when Ω_5 breaks up, we have $M = 0$, that is, the seminvariant bce^3 disappears from the equation, and it is thus a perpetuant: but b^3g^2, b^2df^2, bc^2f^2 and the combinations $c^2de^2 + bce^3$, and $d^5 - bce^3$ are severally reducible.

The degree 15 is evidently the lowest degree for which there is an irreducible quintic seminvariant, and for any higher weight w the number of such seminvariants is equal to the number of capital terms which have the factor D^2EF, viz. this is equal to the number of terms weight $w - 15$ which can be made up with C, D, E, F; and hence

$$\text{for quintic perpetuants G.F. } = x^{15} \div 2 \cdot 3 \cdot 4 \cdot 5.$$

71. For the degree 6, $M = \Pi_6 x \Pi_{15}(x+y)\Pi_{10}(x+y+z)$ is a function of the capitals of the weight 31, and we thence at once infer that

$$\text{for sextic perpetuants G.F. } = x^{31} \div 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6.$$

But it is worth while to write down the expression for M: I do this, annexing to each term the seminvariant (i.e. final term) which belongs to it, arranging these final terms in AO; the value thus arranged is

$M=$		finals in AO
$+1$	D^3E^2FG	$bcei^3$
-2	CD^3EF^2G	$bdehi^2$
$+2$	D^4EF^2G	$befh^3$
-2	CDF^4G	bf^3gh^2
$+1$	F^4G	bg^5
-1	D^4EG^2	c^2di^3
$+1$	CD^4FG^2	cd^2hi^2
$+1$	CD^2EFG^2	$cdegi^2$
-4	$D^4E^2FG^2$	$cdfh^3$
-1	$C^3DF^2G^2$	ceg^3
-1	$D^3F^2G^2$	ce^2h^3
$+4$	$CDEF^2G^2$	$cefgh^2$
$+1$	$C^2F^3G^2$	cf^3h^2
$+4$	EF^3G^2	cfg^4
-1	$C^2D^2G^3$	d^2g^4
$+4$	D^3EG^3	d^2eh^4.

It thus appears that the single sextic perpetuant of the weight 31 is $bcei^3$, and generally that, for any higher weight, the sextic perpetuants are such that the conjugate capital terms contain each of them the factor D^3E^2FG.

The like reasoning shows that

$$\text{for perpetuants of degree } n, \text{ G.F. is } = x^{2^{n-1}-1} \div 2\cdot 3\cdot 4\ldots n.$$

Investigation of the Values of the Foregoing Functions $\Pi_{10}(x+y)$, $\Pi_{15}(x+y)$ *and* $\Pi_{10}(x+y+z)$. *Art. Nos. 72 to 74.*

72. If x, y, z, w, t are the roots of a quintic equation, say

$$\lambda - x\cdot\lambda - y\cdot\lambda - z\cdot\lambda - w\cdot\lambda - t = (1, B, C, D, E, F \mid \lambda, 1)^5 = 0,$$

we require the product $\Pi_{10}(x+y)$ of the sum of two roots in the particular case $B=0$. But in order to the determination of the expression for $\Pi_{15}(x+y+z)$, we require the value of $\Pi_{10}(x+y)$ in the general case, B any value whatever.

Writing

$$x = -\tfrac{1}{2}(\theta + \omega),$$

$$y = -\tfrac{1}{2}(\theta - \omega),$$

and therefore

$$\theta + x + y = 0,$$

we have

$$(\theta + \omega)^5 - 2B(\theta + \omega)^4 + 4C(\theta + \omega)^3 - 8D(\theta + \omega)^2 + 16E(\theta + \omega) - 32F = 0,$$

and the like equation with $-\omega$ for ω. Hence writing $\omega^2 = M$, we have

$$(\theta^5 - 2B\theta^4 + 4C\theta^3 - 8D\theta^2 + 16E\theta - 32F) + M(10\theta^3 - 12B\theta^2 + 12C\theta - 8D) + M^2(5\theta - 2B) = 0,$$

$$(5\theta^4 - 8B\theta^3 + 12C\theta^2 - 16D\theta + 16E) + M(10\theta^2 - 8B\theta + 4C) + M^2 \cdot 1 = 0,$$

which are of the form $A + BM + CM^2 = 0$, $A' + B'M + C'M^2 = 0$, and give therefore by elimination of M the equation

$$-(CA' - C'A)^2 + (BC' - B'C)(AB' - A'B) = 0;$$

the left-hand side is here a function of θ of the degree 10 vanishing when $\theta + x + y = 0$, and which must therefore be, save as to a numerical factor, the product $\Pi_{10}(\theta + x + y)$. And we thus find

$$\Pi_{10}(\theta + x + y) = -48B\theta^9 + 56C\theta^8 + \{-120D - 8BC\}\theta^7 + \{128CE - 128D^2\}\theta^6 + \{-128DE\}\theta^5 + \dots$$

which is

$$= 1024\theta^{10} + \dots + 1024(-F^2 + CDF + 2BEF - BC^2F - D^2E + BCDE),$$

and which therefore for $B = 0$ gives

$$\Pi_{10}(x + y) = -F^2 + CDF - D^2E.$$

73. Suppose now x, y, z, w, t, u are the roots of a sextic equation, say

$$\lambda - x \cdot \lambda - y \cdot \lambda - z \cdot \lambda - w \cdot \lambda - t \cdot \lambda - u = (1, B, C, D, E, F, G \,|\, \lambda, 1)^6 = 0.$$

Considering here the product $\Pi_{20}(x + y + z)$ of the sums of 3 roots, if $B = 0$, this will be a perfect square (for each sum $x + y + z$ is equal to $-$ a sum $(w + t + u)$) say it is the square of $\Pi_{10}(x + y + z)$, where the $x + y + z$ refers to the ten sums each containing x, and we wish to find this function $\Pi_{10}(x + y + z)$. Writing for the equation whose roots are y, z, w, t, u,

$$\lambda - y \cdot \lambda - z \cdot \lambda - w \cdot \lambda - t \cdot \lambda - u = (1, B', C', D', E', F' \,|\, \lambda, 1)^5,$$

we have by what precedes $\Pi_{10}(x+y+z)$ = a function $(*\,|\theta, 1)^{10}$, viz. this is the above-mentioned function with B', C', D', E', F' in place of the unaccented letters. Introducing a new root x and for λ writing as we may do θ, we have

$$\theta - x \cdot \theta - y \cdot \theta - z \cdot \theta - w \cdot \theta - t \cdot \theta - u = (\theta - x) \cdot (1, B', C', D', E', F' \,|\theta, 1)^5,$$

that is, we have

$$(1, B, C, D, E, F, G\,|\theta, 1)^6,$$

$$\begin{array}{ll} B = B' - \theta \text{ or conversely,} & B' = B + \theta, \\ C = C' - B'\theta & C' = C + B\theta + \theta^2, \\ D = D' - C'\theta & D' = D + C\theta + B\theta^2 + \theta^3, \\ E = E' - D'\theta & E' = E + D\theta + C\theta^2 + B\theta^3 + \theta^4, \\ F = F' - E'\theta & F' = F + E\theta + D\theta^2 + C\theta^3 + B\theta^4 + \theta^5, \ = -\frac{G}{\theta}, \\ G = -F'\theta, & \end{array}$$

where I have retained B, but the value hereof is in fact $= 0$. In the foregoing function $(*\,|\theta, 1)^{10}$ with the accented letters, writing for these their values $B' = \theta$, $C' = C + \theta^2$, $D' = D + C\theta + \theta^3$, &c., which belong to $B = 0$, we find

$$1024\Pi_{10}(\theta + y + z) = -(48\theta^5 + 56C\theta^3 + 24D\theta^2 + 64E\theta + 32F)^2$$

$$+(16\theta^3 + 8C\theta + D)\{144\theta^7 + 264C\theta^5 + 72D\theta^4 + 128(C^2 + E)\theta^3 + 192F\theta^2 + 128CE\theta + \ldots\}$$

which equation divides by 64. Writing herein $\theta = x$, we have

$$16\Pi_{10}(x + y + z) = -(6x^5 + 7Cx^3 + 3Dx^2 + 8Ex + 4F)^2$$

$$+(2x^3 + Cx + D)\{18x^7 + 33Cx^5 + 9Dx^4 + 16(C^2 + E)x^3 + 24Fx^2 + 16CEx + 16(CF - DE)\},$$

where $x^6 + Cx^4 + Dx^3 + Ex^2 + Fx + G$ is $= 0$: the value ought, in virtue of this equation, to reduce itself to a mere function of the coefficients, and we in fact find that the equation is

$$16\Pi_{10}(x + y + z) = (16C^2 - 64E)(x^6 + Cx^4 + Dx^3 + Ex^2 + Fx) + 16CDF - 16D^2E - 16F^2,$$

reducing itself to

$$-(16C^2 - 64E)G + 16CDF - 16D^2E - 16F^2,$$

viz. dividing each side by 16, we have

$$\Pi_{10}(x + y + z) = 4EG - C^2G - F^2 + CDF - D^2E,$$

which is the required result. The equation $(\theta^2 - 1)^3 = 0$, for which

$$x, y, z, w, t, u = 1,\ 1,\ 1,\ -1,\ -1,\ -1,$$

gives a numerical verification.

74. I find also, for the same value $B = 0$, the function $\Pi_{15}(x + y)$. Writing, as before,

$$x = -\tfrac{1}{2}(\theta + \omega),$$

and therefore

$$\theta + x + y = 0,$$

we have

$$(\theta+\omega)^6 + 4C(\theta+\omega)^4 - 8D(\theta+\omega)^3 + 16E(\theta+\omega)^2 - 32F(\theta+\omega) + 64G = 0,$$

and the like equation with $-\omega$ for ω. Hence writing $\omega^2 = M$, we have

$$(\theta^6 + 4C\theta^4 - 8D\theta^3 + 16E\theta^2 - 32F\theta + 64G) + M(15\theta^4 + 24C\theta^2 - 24D\theta + 16E) + \ldots = 0,$$

$$(6\theta^5 + 16C\theta^3 - 24D\theta^2 + 32E\theta - 32F) + M(20\theta^3 + 16C\theta - 8D) + M^2 \cdot 6\theta = 0,$$

say these equations are $aM^3 + bM^2 + cM + d = 0$, $pM^2 + qM + r = 0$. Eliminating M, we have

$$\begin{array}{ll}
a^2 \cdot r^3 & a = 1, \\
-ab \cdot qr^2 & b = 15\theta^2 + 4C, \\
 & c = 15\theta^4 + 24C\theta^2 - 24D\theta + 16E, \\
+ac(-2pr^2 + q^2r) & d = \theta^6 + 4C\theta^4 - 8D\theta^3 + 16E\theta^2 - 32F\theta + 64G, \\
+b^2 \cdot pr^2 & \\
 & p = 6\theta, \\
+ad(3pqr - q^3) & q = 20\theta^3 + 16C\theta - 8D, \\
+bc(-pqr) & r = 6\theta^5 + 16C\theta^3 - 24D\theta^2 + 32E\theta - 32F, \\
+bd(-2p^2r + pq^2) & \\
+c^2 \cdot p^2r & \\
-cd \cdot p^2q & \\
+d^2 \cdot p^3 = U. &
\end{array}$$

The equation, as far as I have calculated it, is

$$-32768\theta^{15} - \ldots - 32768(-D^3G + F^3 - CDF^2 + D^2EF) = 0:$$

the left-hand side is here $= -32768\Pi_{15}(x+y)$; and we have therefore

$$\Pi_{15}(x+y) = -D^3G + F^3 - CDF^2 + D^2EF,$$

the required result. It may be remarked that, writing $G = 0$ and throwing out a factor $-F$, we have $-F^2 + CDF - D^2E$, which is the expression for $\Pi_{10}(x+y)$ in the quintic equation.

We have

$$\Pi_6 x \Pi_{15}(x+y)\Pi_{10}(x+y+z)$$

$$= G\{-D^3G + (F^2 - CDF + D^2E)F\}\{(4E - C^2)G - F^2 + CDF - D^2E\}.$$

the developed expression whereof is the foregoing value

$$M = D^3E^2FG - 2CD^3EF^2G + \&c., \text{ ante No. 71.}$$

The Operators $P - \delta b$ and $Q - 2\omega b$. Art. Nos. 75 to 84.

75. The analogous theory for non-unitariants is established, *ante* Nos. 24 *et seq.* For seminvariants, we have

$$P = b\partial_a + c\partial_b + d\partial_c + \ldots,$$

$$Q = c\partial_b + 2d\partial_c + \ldots,$$

or more definitely, if the seminvariant operated upon be of the degree δ, the weight ω and extent σ, say its highest letter is a_σ, $= p$, then

$$P = b\partial_a + c\partial_b + d\partial_c + \ldots + q\partial_p,$$

$$Q = c\partial_b + 2d\partial_c + \ldots + \sigma q\partial_p,$$

then we have

$$P - \delta b, \quad Q - 2\omega b,$$

operators each of them of the deg. weight 1.1, viz. each of them operating upon a seminvariant S of the deg. weight $\delta.\omega$ gives a seminvariant S' of the deg. weight $\delta + 1.\omega + 1$; moreover, a new letter q is introduced, or say the extent is increased from σ to $\sigma + 1$. For the proof, it is only necessary to show that $\Delta(P - \delta b)S$ and $\Delta(Q - 2\omega b)S$ are each $= 0$, but it is unnecessary to do this, as the like proof has already been given for non-unitariants.

The two seminvariant operators were first considered in my paper "On a theorem relating to seminvariants," *Quart. Math. Journ.* t. XX. (1885), pp. 212, 213, [844].

76. We may, instead of $P - \delta b$ and $Q - 2\omega b$, consider the linear combination

$$Y = 2\omega(P - \delta b) - \delta(Q - 2\omega b), \text{ that is, } 2\omega P - \delta Q,$$

which is of deg. weight 0.1, viz. it leaves the degree unaltered, while increasing as before the weight, and also the extent, each by unity. And again, the combination

$$Z = \sigma(P - \delta b) - (Q - 2\omega b),$$

that is,

$$Z = \sigma P - Q - (\sigma\delta - 2\omega)b,$$

where observe that $\sigma P - Q$, $= \sigma b\partial_a + (\sigma - 1)c\partial_b + \ldots + 1 \cdot p\partial_q$ does not contain the new letter q; the operator Z is thus of the deg. weight 1.1 increasing the degree and also the weight each by unity, but leaving the extent unaltered.

There is a special case which it is important to attend to, we may have $\sigma\delta - 2\omega = 0$, viz. this is the case when the seminvariant operated upon is in regard to the letters comprised therein an invariant. Here the two combinations Y, Z are equivalent to each other, each of them is $= \sigma b\partial_a + (\sigma - 1)c\partial_b + \ldots + 1 \cdot p\partial_q$, which is an annihilator of the seminvariant (invariant) operated upon. Hence in this case we cannot replace the original forms by the linear combinations, but must retain one (no matter which) of the original forms $P - \delta b$, $Q - 2\omega b$.

77. We can, by means of the foregoing operators, starting from the quadric seminvariants $c - b^2$, &c., derive in order the seminvariants for the successive weights $3, 4, 5, \ldots$.

Thus writing down the series of finals (in AO as before),

$$\begin{array}{ccccccc} b^2, & b^3, & c^2, & bc^2, & d^2, & & \dots \\ & b^4, & & b^4c, & bd^2, & & e^2, \\ & & b^5, & & b^5c, & & b^2c^2 \\ & & & & & b^3c^2,\ b^4d^2,\ bc^3,\ cd^2 & \end{array}$$

I proceed as follows, observing, however, that when the function operated upon is an invariant seminvariant we must instead of Z write $P - \delta b$.

$$\begin{array}{l|l|l|l|l} b^2 \text{ emerges,} & b^3 = Zb^2 & c^2 \text{ emerges,} & bc^2 = Zc^2 & d^2 \text{ emerges,} \\ & b^4 = Zb^3 & b^5 = Zb^4 & c^3 = Ybc^2 & bc^3 = Zc^3 \\ & & & b^2c^2 = Zbc^2 & b^3c^2 = Zb^2c^2 \\ & & & b^3 = Zb^2 & b^7 = Zb^6 \\ & & & & c^4 = Ybc^3 \\ & & & & b^2c^3 = Zbc^2 \end{array}$$

viz. whenever the seminvariant to be obtained has a final containing b, it is obtained by means of the operator Z (or it may be $P - \delta b$), but when there is no b then by the operator Y.

The seminvariants operated upon may be blunt or sharp, but there is an advantage in operating on the sharp forms as these are more simple, and we thereby obtain for the next superior weight forms more nearly approximating to the sharp forms. We do not however by thus operating on a sharp form obtain directly a sharp form; to do this, the form obtained must be modified by adding thereto a numerical multiple or multiples of a preceding sharp form: and thus the theory does not determine beforehand the forms of the sharp seminvariants. But making at each step the necessary modification (if any) we have thereby, when the sharp seminvariants of the next preceding weight are known, a very convenient process for the calculation of the sharp seminvariants of any given weight, in the AO arrangement of their final terms. Thus for the weight 10; $k\infty f^2$ is taken to be known, the next two forms $ci\infty ce^2$ and $dh\infty b^2e^2$ are calculated each from $j\infty bc^2$, the expression for which is

$$= j - 9bi + 20ch - 28dg + 14ef + 16b^2h - 56bcg + 112bdf - \dots$$

We have for $j\infty bc^2$, $\delta = 3$, $\omega = 9$, $\sigma = 9$: and therefore

$$\tfrac{1}{3}Y = 6b\partial_a + 5c\partial_b + 4d\partial_c + 3e\partial_d + 2f\partial_e + g\partial_f - i\partial_h - 2j\partial_i - 3b,$$

$$Z = 9b\partial_a + 8c\partial_b + 7d\partial_c + 6e\partial_d + 5f\partial_e + 4g\partial_f + 3h\partial_g + 2i\partial_h + j\partial_i - 9b.$$

78. I exhibit the calculation as follows:

$$\tfrac{1}{3}Y(j\infty bc^2)$$

[Table: ten-column numerical worksheet exhibiting the operation $\frac{1}{3}Y$ on the seminvariant $j\infty bc^2$. The rows are labelled by the monomials in the result (k, bj, ci, dh, eg, f^2, b^4h, bch, bdg, bef, b^2cg, cdf, c^2g, b^3df, ce^2); the columns are headed $1, 2, \ldots, 9$ corresponding to the nine terms $6b\partial_a, 5c\partial_b, \ldots$ of $\frac{1}{3}Y$, with two further columns at the right marked with a dagger † and an asterism $*$ holding the additional term added to fix the initial coefficient and the final tabulated result for $ci\infty ce^2$ respectively. Column totals are noted at the bottom; the column $\div 70$ records the divisor applied to obtain the final tabulated form.]

$$Z(j\infty bc^2)$$

[Table: ten-column numerical worksheet exhibiting the operation Z on the seminvariant $j\infty bc^2$. The rows are labelled by the monomials in the result (k, bj, ci, dh, eg, f^2, b^4h, bch, bdg, bef, b^2cg, cdf, c^2g, b^3df, ce^2); the columns are headed $1, 2, \ldots, 10$ corresponding to the ten terms $9b\partial_a, 8c\partial_b, \ldots$ of Z, with two further columns at the right marked with a dagger † and an asterism $*$ holding the additional term $+32(ci\infty ce^2)$ added to fix the initial coefficient and the final tabulated result for $dh\infty b^2e^2$ respectively. Column totals are noted at the bottom; the column $\div -18$ records the divisor applied to obtain the final tabulated form.]

The numbers $(1, 2, \ldots, 9)$ and $(1, 2, \ldots, 10)$ at the head of the columns refer to the nine terms $6b\partial_a, 5c\partial_b, \ldots$ of $\frac{1}{3}Y$, and the ten terms $9b\partial_a, 8c\partial_b, \ldots$ of Z respectively, these several operations being performed on $(j\infty bc^2)$ the value of which is given above: the daggers † denote the additions which have to be made in order to obtain the proper initial term, viz. for the first † the added term is $+3(k\infty f^2)$ and for the second † the added term is $+32(ci\infty ce^2)$: the headings $\div 70$ and $\div -18$ explain themselves, and the columns headed with an asterism $*$ give the results, viz. the first of these is $(ci\infty ce^2)$ and the second of them is $(dh\infty b^2e^2)$. As appears above, the value of the first of these is used in the second † column for obtaining that of the second of them.

79. We may operate with $P - \delta b$ and $Q - 2\omega b$ on a product (deg. weight $\delta.\omega$) ST of two seminvariants S, T, deg. weights $\delta'.\omega'$ and $\delta''.\omega''$ respectively, $\delta = \delta' + \delta''$, $\omega = \omega' + \omega''$. We have

$$(P - \delta b)ST = S \cdot PT + T \cdot PS - (\delta' + \delta'')bST = S(P - \delta''b)T + T(P - \delta'b)S,$$

where $(P - \delta'b)S$ and $(P - \delta''b)T$ are each of them a seminvariant. And similarly,

$$(Q - 2\omega b)ST = S(Q - 2\omega''b)T + T(Q - 2\omega'b)S,$$

where $(Q - 2\omega'b)S$ and $(Q - 2\omega''b)T$ are each of them a seminvariant. That is, operating either with $P - \delta b$ or $Q - 2\omega b$ on a product, we have a sum of products; and therefore also operating upon a sum of products (each product being of the deg. weight $\omega.\delta$) we have a sum of products, each product in such sum being of the deg. weight $\omega + 1.\delta + 1$, and moreover of the extent $\sigma + 1$. And instead of binary products, we may, it is clear, consider ternary, quaternary, &c., products.

The like theorem applies to the derived operators Y and Z, but as to Y there is a specialty to be noticed. We have

$$\begin{aligned} Y \cdot ST &= 2\omega(P - \delta b)ST - \delta(Q - 2\omega b)ST, \\ &= 2\omega\{S(P - \delta''b)T + T(P - \delta'b)S\} - \delta\{S(Q - 2\omega''b)T + T(Q - 2\omega'b)S\}, \\ &= S\{2\omega(P - \delta''b)T - \delta(Q - 2\omega''b)T\} + T\{2\omega(P - \delta'b)T - \delta(Q - 2\omega'b)S\}, \end{aligned}$$

where the whole of the right-hand side as being equal to $Y.ST$ is of the degree δ, but except in the particular case $\left(\frac{\delta}{\omega} = \frac{\delta'}{\omega'} = \frac{\delta''}{\omega''}\right)$ the separate products $S\{\ldots\}$ and $T\{\ldots\}$ which occur on the right-hand side are each of them of the degree $\delta + 1$.

It is scarcely necessary, but it may be proper, to remark that we frequently combine by addition a seminvariant S of the deg. weight $\delta.\omega$ with a seminvariant T deg. weight $\delta - \epsilon.\omega$ of the same weight but of an inferior degree, but when this is done we regard the T as standing for $a^\epsilon T$, and as being thus of the same deg. weight $\delta.\omega$. We have

$$(P - \delta b)a^\epsilon T = a^\epsilon PT + TPa^\epsilon - (\epsilon + \delta - \epsilon)ba^\epsilon T, \; = a^\epsilon\{P - (\delta - \epsilon)b\}T + T(P - b)a^\epsilon,$$

where $(P-\epsilon b)a^\epsilon = (\epsilon-\epsilon)b = 0$, and consequently $(P-b\delta)a^\epsilon T = \{P-(\delta-\epsilon)b\}T$; viz. for the operation upon T, we regard $P-\delta b$ as standing for $P-(\delta-\epsilon)b$. As regards Q, we have $(Q-2\omega b)a^\epsilon T = (Q-2\omega b)T$; viz. the degree of T does not here present itself.

80. We may write

$$(2\omega P - \delta Q)S = S',$$

the new seminvariant S' being of the weight $\omega+1$; hence also

$$\{(2\omega+2)P-\delta Q\}\cdot\{2\omega P-\delta Q\}S = S'',$$

where S'' is of the weight $\omega+2$; viz. we have an operator

$$\{(2\omega+2)P-\delta Q\}\{2\omega P-\delta Q\}$$

which, operating on a seminvariant of the deg. weight $\delta.\omega$, gives a seminvariant of the deg. weight $\delta.\omega+2$. This is

$$(4\omega^2+4\omega)P^2-(4\omega+2)\delta PQ-2\omega\delta(Q\cdot P)+\delta^2Q^2+(4\omega^2+4\omega)P\cdot P-(4\omega+2)\delta P\cdot Q+\delta^2 Q\cdot Q,$$

where P^2, PQ, QP and Q^2 are the mere algebraical squares and products, while $P\cdot Q$ and $Q\cdot P$ denote respectively P operating on Q and Q operating on P; and since $PQ=QP$, this is

$$(4\omega^2+4\omega)P^2-(4\omega+2)\delta PQ+\delta^2Q^2+(4\omega^2+4\omega)P\cdot P-(2\omega\delta+2\omega)\delta P\cdot Q+\delta^2Q\cdot Q.$$

Recollecting that

$$P = b\partial_a + c\partial_b + d\partial_c + \ldots,\quad Q = c\partial_b + 2d\partial_c + \ldots,$$

we have

$$P\cdot P = c\partial_a + d\partial_b + e\partial_c + \ldots,$$

$$P\cdot Q = d\partial_b + 2e\partial_c + \ldots,$$

$$Q\cdot P = c\partial_a + 2d\partial_b + 3e\partial_c + \ldots,\ = P\cdot P + P\cdot Q,$$

$$Q\cdot Q = 1\cdot 2d\partial_b + 2\cdot 3e\partial_c + \ldots,$$

and attending to the relation just obtained $Q\cdot P = P\cdot P + P\cdot Q$, we find that the operator may be written

$$(4\omega^2+4\omega)\{P^2-(\delta-1)P\cdot P\}$$

$$-(4\omega+2)\delta\{PQ-\omega P\cdot P-\tfrac{1}{2}(\delta-3)P\cdot Q\}$$

$$+\delta^2\{Q^2+Q\cdot Q-\tfrac{1}{2}(4\omega+2)P\cdot Q\};$$

in fact, here the terms in P^2, PQ, Q^2 are in the original form P, Q, while those in $P\cdot P$, $P\cdot Q$, $Q\cdot Q$ are

$$(4\omega^2+4\omega)(1-\delta)P\cdot P+(4\omega^2+2\omega)\delta P\cdot P-\tfrac{1}{2}(4\omega+2)(\delta^2-3\delta)P\cdot Q+\delta^2Q\cdot Q$$

which are

$$= (4\omega^2+4\omega-2\omega\delta)P\cdot P-(4\omega+2)\delta P\cdot Q+\delta^2Q\cdot Q,$$

agreeing with the original form

$$(4\omega^2+4\omega)P\cdot P-(2\omega+2)\delta P\cdot Q-2\omega\delta(P\cdot P+P\cdot Q)+\delta^2Q\cdot Q.$$

81. I find that each of the three parts is separately an operator, viz. that we have

$$P^2 - (\delta - 1)P \cdot P,$$

$$PQ - \omega P \cdot P - \tfrac{1}{2}(\delta - 3)P \cdot Q,$$

$$Q^2 + Q \cdot Q - \tfrac{1}{2}(4\omega + 2)P \cdot Q,$$

each of them an operator which, operating on a seminvariant of deg. weight $\delta.\omega$, gives a seminvariant of deg. weight $\delta.\omega + 2$.

I verify this for the first of the three operators, say

$$\Omega = P^2 - (\delta - 1)P \cdot P = P^2 + P \cdot P - \delta\Theta,$$

if for a moment

$$\Theta = P \cdot P, \ = c\partial_a + d\partial_b + \ldots$$

is put $= \Theta$.

Here for a seminvariant S, we have

$$\Omega S = (aP - b\delta)S = P(PS) - \delta\Theta S.$$

Writing $S' = (aP - b\delta)S$, then S' is a seminvariant, degree $= \delta + 1$, and then if $S'' = (aP - b(\delta + 1))S'$, S'' is a seminvariant, degree $= \delta + 2$. We have $PS = a^{-1}(S' + b\delta S)$, and thence

$$\Omega S = Pa^{-1}(S' + b\delta S) - \delta\Theta S, \ = -b(S' + b\delta S) + P(S' + b\delta S) - \delta\Theta S.$$

Here

$$P(S' + b\delta S) = PS' + b\delta PS + \delta S \cdot Pb = PS' + b\delta a^{-1}(S' + b\delta S) + \delta S \cdot c,$$

and hence

$$\Omega S = S'' + 2b\delta S' + \{c\delta + b^2(\delta^2 - \delta)\}S - \delta\Theta S.$$

This will be a seminvariant if $\Delta \cdot \Omega S = 0$; we have

$$\Delta \cdot \Omega S = \Delta S'' + 2b\delta\Delta S' + \{c\delta + b^2(\delta^2 - \delta)\}\Delta S - \delta(\Delta\Theta + \Delta \cdot \Theta)S + \{2\delta S' + 2b(2b\delta^2 - 2b\delta)\}\Delta S,$$

or, omitting the terms in $\Delta S''$, $\Delta S'$, ΔS which respectively vanish, this is

$$= -\delta(\Delta\Theta + \Delta \cdot \Theta)S.$$

But since $PS = S' + b\delta S$, and from $\Delta\Omega = 0$ we deduce $0 = (\Theta\Delta + \Theta \cdot \Delta)S$, the equation becomes

$$\Delta \cdot \Omega S = -\delta(2\Theta\Delta + 2\Theta\Delta)S,$$

and from

$$\Delta = a\partial_b + 2b\partial_c + 3c\partial_d + \ldots$$

we have

$$\Theta = c\partial_a + d\partial_b + e\partial_c + \ldots,$$

$$\Delta \cdot \Theta = 2b\partial_a + 3c\partial_b + 4d\partial_c + \ldots,$$

$$\Theta \cdot \Delta = c\partial_a + 2d\partial_b + \ldots,$$

and thence

$$\Delta \cdot \Theta - \Theta \cdot \Delta = 2b\partial_a + 2c\partial_b + 2d\partial_c + \dots, \; = 2P,$$

and we have thus the required equation $\Delta \cdot \Omega S = 0$.

82. If instead of P, Θ, we write B, C, so that

$$B = b\partial_a + c\partial_b + d\partial_c + \dots,$$

$$C = B \cdot B = c\partial_a + d\partial_b + e\partial_c + \dots,$$

and put further

$$D = B \cdot C = d\partial_a + e\partial_b + f\partial_c + \dots,$$

$$E = B \cdot D = e\partial_a + f\partial_b + g\partial_c + \dots,$$

then the first of the foregoing operators is $B^2 - (\delta - 1)C$, or reversing the sign, say it is $(\delta - 1)C - B^2$, which is the first of a series of operators

$$(\delta - 1)C - B^2,$$

$$(\delta - 1)(\delta - 2)D - 3(\delta - 2)BC + 2B^3,$$

$$(\delta - 1)(\delta - 2)(\delta - 3)E - 4(\delta - 2)(\delta - 3)BD + 3(\delta - 3)BC^2 - 3B^2C,$$

which are of the deg. weights $0.2, 0.3, 0.4$, &c., respectively, viz. operating upon a seminvariant of deg. weight $\delta.\omega$ they leave the degree unaltered, but increase the weight by 2, 3, 4, ... respectively.

It is to be observed that B^2, BC, B^3, &c., denote the mere algebraical powers and products of the symbols B, C, D, &c., without any operation of one symbol on another.

As a simple illustration, take $(C - B^2)(ac - b^2)$: here,

$$(C - B^2)(ac - b^2) = (-2bd + c^2) - (-2bd + c^2) = e - 4bd + 3c^2;$$

$$\text{Value is } e - 4bd + 3c^2;$$

and similarly for $(C - B^2)(ae - 4bd + 3c^2)$, here:

$$(C - B^2)(ae - 4bd + 3c^2) = g - 4bf + (6 + 1)ce - \dots$$

$$= -2bf + 8ce - \dots$$

$$\text{Value is } g - 6bf + 15ce - 10d^2.$$

A direct proof may of course be obtained for any one of the foregoing operators; viz. calling it Ω, it may be shown that $\Delta\Omega S = 0$. I have not considered the like question of the derivation of series of operators from the other two forms

$$PQ - \omega P \cdot P - \tfrac{1}{2}(\delta - 3)P \cdot Q \quad \text{and} \quad Q^2 + Q \cdot Q - \tfrac{1}{2}(4\omega + 2)P \cdot Q$$

respectively.